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Prevalence of Dermatoses among Pregnant Women
in White Nile State 2023-2024

A research thesis submitted in partial fulfillment of the degree of
Bachelor of Medicine and Surgery

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Dedication

This research is dedicated for our families, our friends and our Supervisor, whom unwavering support have been the cornerstone in this journey. To the resilient women of Sudan, we sincerely dedicate this study to your courage in the face of adversity.

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Abbreviations

Abbreviation	The Meaning/Synonyms
PUPPP	Pruritic Urticarial Papules and Plaques of Pregnancy/ Polymorphic eruption of pregnancy
ICP	Intrahepatic Cholestasis of Pregnancy
PP	Prurigo of pregnancy
PF	Pruritic folliculitis of pregnancy

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Abstract

Introduction:

women's bodies undergo numerous and significant changes throughout pregnancy in many body systems including the skin affecting it in a variety of ways from physiological alteration to pathological one, our study determine the prevalence and patterns of the skin changes affecting pregnant women's bodies .

Materials and Methods:

Across-sectional descriptive study was conducted ,and data was collected from 100 pregnant women in White Nile State through a detailed self-administered questionnaires.

Study results :

The prevalence of skin changes was 69% and the most common was itching(23.9%) affecting mainly the second trimester (50.7%),and age group between 19-40 years old.

Only 30.1% of participants had a previous history of skin changes and 52.2% reported no effect on their quality of life.

Conclusion:

Significant proportion of women experience variable skin changes during pregnancy , the age distribution of participants aligns with the prevalence of skin changes is higher comparing to some studies, with the second trimester was most commonly associated with the beginning of skin changes.

CHAPTER ONE

1.1 Introduction:

1.1.1 Background

Women's bodies undergo numerous and significant changes throughout pregnancy in terms of their endocrine, metabolic, immune and circulatory systems. These circumstances can affect the skin in a variety of ways, from physiological alterations to pathologic eruptions that are specifically related to pregnancy^[1,2].

Physiological changes to the skin may occur during pregnancy.

But while some dermatoses are exclusive to pregnancy, others are impacted by it. During pregnancy, the body experiences these changes as a result of the interaction of several factors^[3,4].

The majority of pregnant women have some degree of hyperpigmentation. The most usually affected areas are the face, areolae, axillae, and genitalia. Nearly 90% of women have linea nigra. Melasma, often known as chloasma or the pregnant mask, is the second-most. As a result of physiological hormonal changes during pregnancy, benign skin changes such as striae gravidarum, hyperpigmentation, and hair, nail, and common can also happen. Stretch marks, or striae gravidarum, are most common in the third trimester in women. Bands that mostly affect the abdomen but can also affect the buttocks, breasts, thighs, or arms. They appear as pink-purple, atrophic bands^[5-8].

Pregnancy may cause pre-existing skin disorders like dermatitis, psoriasis, any candidal infections, or fungal infections to worsen.

The majorities of the inflammatory skin dermatoses that are particular to pregnancy are benign and go away in the postpartum period^[8].

The first ever classification for pregnancy specific dermatoses was given in 1983 by Holmes and Black which comprised the skin lesions into four^[5]:

- Pemphigoid gestationis (Herpes gestationis)
- Polymorphic eruption of pregnancy (PUPPP)
- Prurigo of pregnancy (PP) and
- Pruritic folliculitis of pregnancy (PF).

Another classification given in 1998 by Shornick, included PG, PEP and PP with intrahepatic cholestasis of pregnancy (ICP) in addition. The most recent classification was given in 2006 by Ambros-Rudolph et al; they added "atopic

eruption of pregnancy (AEP)” in the pre-existing classification ^[7-9]. They subdivided AEP into eczema in pregnancy (ep), prurigo of pregnancy and pruritic folliculitis of pregnancy. So as per the new classification four main skin conditions present in pregnancy are:

Atopic Eruption of Pregnancy, eczema in pregnancy , prurigo of pregnancy, Pruritic folliculitis of pregnancy , Polymorphic eruption of pregnancy , Pemphigoid gestationis and Intrahepatic cholestasis of pregnancy under dermatoses of pregnancy^[2].

1.2 Literature Review:

Dermatosis during pregnancy encompasses a spectrum of cutaneous conditions that can manifest uniquely during gestation, impacting both maternal health and fetal well-being. This review aims to provide a comprehensive overview of the epidemiology, clinical manifestations, pathophysiology, risk factors, and management strategies associated with dermatoses in pregnancy.

Epidemiology and Prevalence:

Dermatoses in pregnancy are common, affecting up to 90% of pregnant women to varying degrees ^[10-12]. The prevalence and incidence rates of specific dermatoses vary across populations, geographic regions, and ethnicities ^[13, 14]. While some conditions, such as pruritic urticarial papules and plaques of pregnancy (PUPPP), are more prevalent in primiparous women, others, like intrahepatic cholestasis of pregnancy (ICP), show no such predilection ^[15, 16]. Additionally, certain dermatoses, such as pemphigoid gestationis, exhibit ethnic variations, being more common in Caucasian populations ^[5].

Many studies were conducted one of them was done by Lt Col Md Humayun Kabir et al and they reported that about 6.9% pregnant women had specific dermatoses, and 59.48% patients were in third trimester, 37.9% were in between the age of 25 to 29 years, and the most typical dermatoses during pregnancy were pruritic urticarial papules and plaques^[4].

Another study done in South West Nigeria of patients attending antenatal clinics by Olusola Olabisi Ayanlowo et al they found that itching was the most prevalent symptom 42.2%, and dermatoses were present in 85.5% of cases, and 3.7% of pregnant women had atopic eruption of pregnancy with pruritic urticarial papules and plaques^[17].

On the other hand Mariusz Sikora, Ciechanowicz et al found that pregnant women who reported skin changes were 74.78% and 94.8% of cases had physiological skin alterations, Pregnancy-related dermatoses were noted in 14% of cases ^[2].

Clinical Manifestations:

Dermatoses in pregnancy encompass a diverse array of clinical presentations, ranging from benign pruritus to potentially life-threatening conditions. Common dermatoses include PUPPP, which typically presents as pruritic erythematous

papules and plaques in the third trimester, and melasma, characterized by hyperpigmented macules on sun-exposed areas ^[18, 19]. Other notable conditions include ICP, characterized by pruritus and elevated bile acids, and gestational pemphigoid, which manifests as tense bullae and urticarial lesions^{[7][20]}. Additionally, pre-existing dermatological conditions such as psoriasis and eczema may undergo exacerbation or remission during pregnancy. ^[21, 22]

Pathophysiology:

The pathogenesis of dermatoses in pregnancy is multifactorial, involving hormonal, immunological, genetic, and environmental factors. Hormonal changes, particularly elevated estrogen and progesterone levels, play a central role in many pregnancy-related dermatoses by influencing skin physiology and immune function ^[23, 24]. Immunological alterations, including shifts in T-cell subsets and cytokine profiles, contribute to the development of conditions such as pemphigoid gestationis and atopic eruption of pregnancy ^[25, 26]. Genetic predisposition also plays a role, as evidenced by familial clustering observed in conditions like intrahepatic cholestasis of pregnancy ^[27].

Risk Factors and Predictors:

Several maternal and fetal factors have been identified as risk factors for the development of dermatoses during pregnancy. Maternal age, parity, gestational age, and pre-existing skin conditions are among the most commonly reported predictors ^[28, 29]. Advanced maternal age and primiparity are associated with an increased risk of developing certain dermatoses, while multiple gestations may predispose women to conditions such as striae gravidarum and PUPPP ^[30, 31]. Additionally, underlying liver dysfunction and genetic polymorphisms have been implicated in the pathogenesis of intrahepatic cholestasis of pregnancy ^{[11][25]}.

1.3 Problem Statement:

Pregnancy results in cutaneous changes in more than 90% of women ^[32].

Dermatoses unique to pregnancy are important to recognize for the clinician as they carry considerable morbidity for pregnant mothers and in some instances constitute a risk to the fetus.

Besides, women affected with itch-related dermatosis may endure pain, burning and tickling sensations as one study reported a prevalence of itch as high as 38% and 70% of them suffered on a daily basis ^[33].

Available studies show wide variations and conflicting data in the incidence of specific pregnancy dermatoses as it doesn't manifest as unified entity it may appear in form of many conditions for example: psoriasis, stretch marks or commonly as pruritis gravidarum and pruritic eruption of pregnancy ^{[1] [34, 35]}.

1.4 Justification:

In Sudan there is a lack of an exact numerical data about the prevalence of

dermatosis during pregnancy, and obtaining this data will facilitate in the preparation of required measures in the treatment of dermatosis and preventing harmful effects associated with it.

A pregnant woman with a pruritic skin eruption requires immediate evaluation and diagnosis because delayed diagnosis or misdiagnosis may pose significant risk to the fetus and the mother. It is necessary to know how to diagnosis and treat this conditions to establish a better outcome for the mother and the fetus.

1.5 Research Objectives:

1.5.1 General objectives:

This study aims to determine the prevalence and pattern of dermatosis among pregnant women.

1.5.2 Specific objectives:

- 1- To determine the prevalence of specific dermatosis associated with pregnancy.
- 2- To find out the most prevalent skin change during pregnancy.
- 3 - To identify most affected age group.
- 4- To assess the quality of life under the effect of dermatosis.
- 5- To identify and organize trimesters on the bases of incidence of skin changes.

CHAPTER TWO

MATERIALS AND METHODS

2.1 Study design:

Cross-sectional descriptive study.

2.2 Study area

White Nile State, Obstetric and gynecological clinics at Rabak, Kosti and Ad-Dewym cities .

2.3 Study population

Pregnant women regardless of their age or gestational age.

2.3.1 Inclusion criteria:

All pregnant females attending antenatal care clinic.

2.3.2 Exclusion criteria:

1. Refusal to participate in the study.
2. Pregnant women known to have a preceding dermal condition.
3. Pregnant women having associated condition as diabetes, hypertension, hypothyroidism, hyperthyroidism,

2.4 Sample size and technique

Sample size of 100 was calculated using the simple formula (Kish 1965 Formula).

$$n = z^2 p (1-p)/d^2$$

n = sample size

p = proportion of the estimate (obtained from the previous study. [2])

d = margin of error

z = Z score

$$n = (1.96)^2 (74.78) (25.22)/25$$

$$n = 290$$

This number wasn't met during this study due to overwhelming circumstances. The researchers managed to collect a sum of 100 samples.

2.5 Methods of data collection

A self-administered online questionnaire was designed by the authors to collect

the necessary data from participants after acquiring their consent, the questionnaire consists of questions regarding age, gestational age, the status of participants general and dermatological health and the history of gestation and dermatological conditions to meet the objectives of this study.

The data was collected randomly among pregnant women who attend antenatal clinics, and it was analyzed using SPSS version 26.0

2.6 Data analysis

- The data was checked and validated for analysis .
- The data was analyzed by analyst using SPSS version 25.
- Variables were demonstrated as frequency tables and figures.
- Results were discussed and compared with many correlated studies .

2.7 Ethical Considerations

Benefits of the study:

This study would be beneficial as it reflect an issue that affect the life of women during pregnancy. We anticipate that the findings on this study will prompt better consideration for dermatosis among medical practitioners locally and globally.

Ethical clearance:

1. No risks or discomforts were inteicipated from taking part in this study. Participants can discontinue filling the questionnaire at any time.
2. Voluntary informed consentswere taken from participants prior to filling the questionnaire.
3. A permission from Community medicine and Research Department at Alzaeim Alazhari University, Faculty of Medicine, was Obtained to conduct this study.
4. A permission from Khartoum State Ministry of Health was obtained to conduct this study.

CHAPTER THREE

STUDY RESULTS

3- Study Results

The study aimed to determine the prevalence and pattern of dermatosis among pregnant women in the White Nile State. A total of 100 participants were included in the study, with varying demographic and obstetric characteristics.

The age distribution among the participants showed that the majority (45.0%) fell within the 19-29 age group, followed by the 30-40 age group (40.0%). A smaller percentage of participants were in the 41-50 age group (6.0%), while 9.0% were less than 18 years old.[Figure 1]

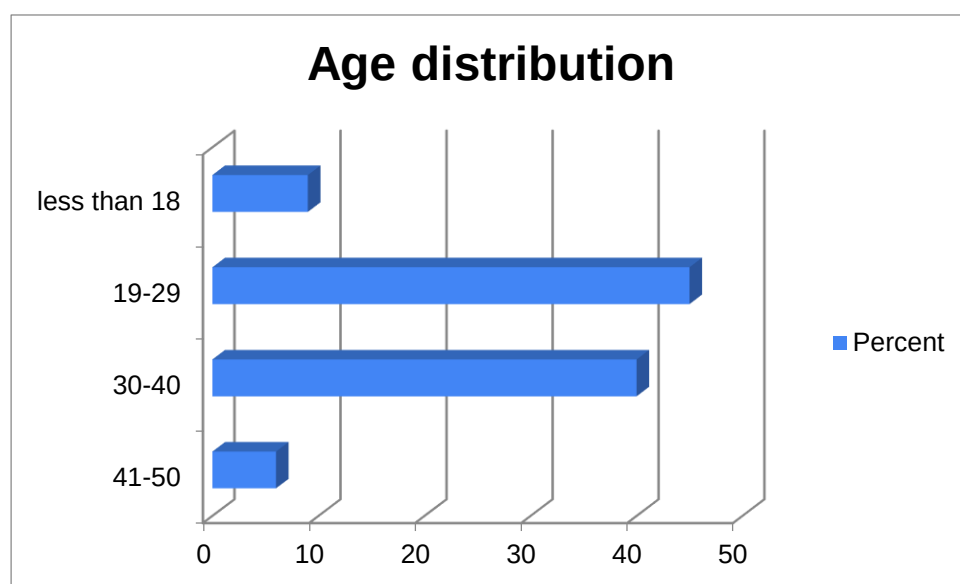


Figure 1 : **Age Distribution among participants**

Regarding the history of previous pregnancies, 72.4% of the participants reported having previous pregnancies, while 27.6% had no history of previous pregnancy. Among those (N=72), the distribution was as follows: more than three pregnancies (26.4%), abortion (1.4%), three pregnancies (16.7%), two pregnancies (25.0%), and one pregnancy (30.6%). [Figure 2][Table1]

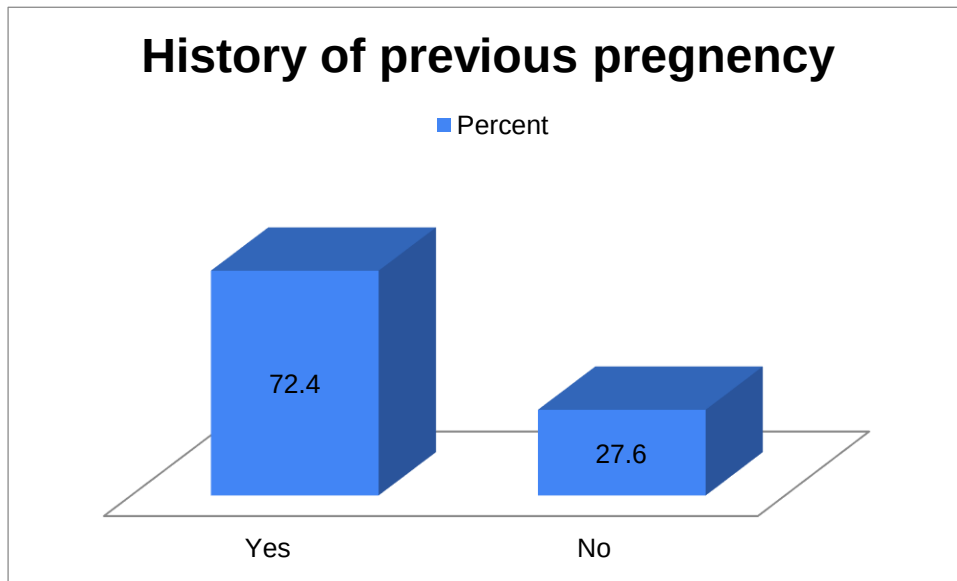


Figure 2: History of previous pregnancy

Table 1: Obstetric History among participants.

Obstetric History	Frequency	Percent
More than three	19	26.4
Abortion	1	1.4
Three	12	16.7
Two	18	25.0
One	22	30.6
Total	72	100.0

The presence of skin changes among the participants was reported by 14.0% of the participants, while 86.0% did not have any skin changes. [Figure 3] Among the 15 participants who reported having a skin disease, the distribution of specific types of dermatosis was as follows: psoriasis (20.0%), allergy (13.3%), alopecia (6.7%), vitiligo (13.3%), stretching marks (6.7%), eczema (26.7%), and varicose veins (6.7%). It is worth noting that 6.7% of participants reported that they did not have any existing skin diseases.[Table 2]

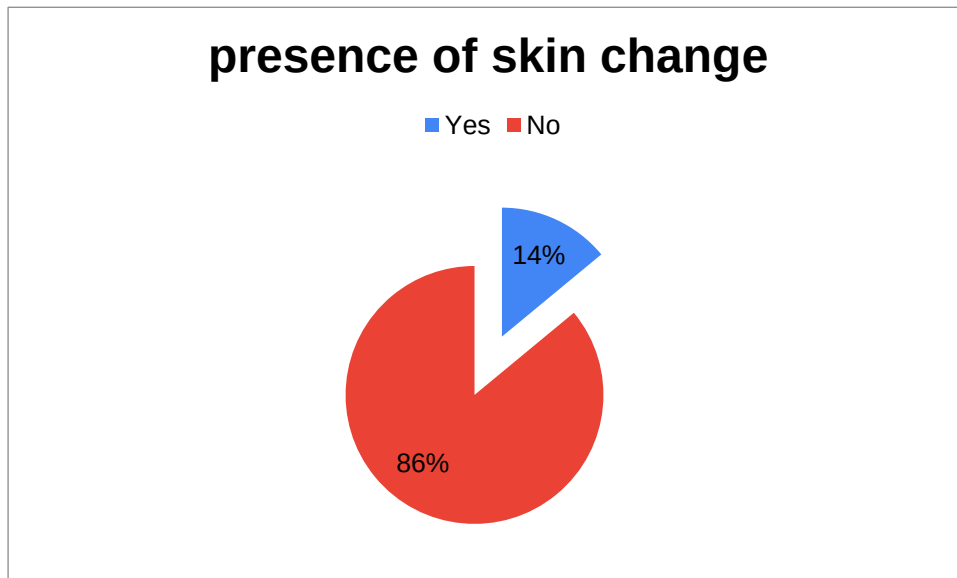


Figure 3: **Presence of skin changes among participants.**

Table 2: Type of skin disease among participants

Type of skin disease	Frequency	Percent
Does not exist	1	6.7
Psoriasis	3	20.0
Allergy	2	13.3
Alopecia	1	6.7
Vitiligo	2	13.3
Stretching mark	1	6.7
Eczema	4	26.7
varicose veins	1	6.7

Total	15	100.0
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Of the total participants, 69.0% reported experiencing skin changes during pregnancy, while 31.0% did not. Among the 67 participants who described their skin changes, the most frequently reported skin changes included itching (23.9%), back spots (22.4%), acne (4.5%), and stretch marks (4.5%), among others.[Table 3,4]

Table 3: Presence of skin changeduring pregnancy among participants

Presence of skin changeduring pregnancy	Frequency	Percent
Yes	69	69.0
No	31	31.0
Total	100	100.0

Table 4: Description of the skin changes type among participants

Description of the skin changes	Frequency	Percent
Blisters	1	1.5
Rash	4	6.0
Itching rash	2	3.0
Itching skins line	1	1.5
Itching Acne	1	1.5
Itching black spot	4	6.0
Itching Redness	3	4.5
Itching	16	23.9
Blind Pimple	1	1.5

Light colored spots	1	1.5
Black rash	2	3.0
melasma	1	1.5
Back spot	15	22.4
Redness	3	4.5
stretch markand lineanigra	3	4.5
Stretch mark	3	4.5
Linea nigra	3	4.5
Acne	3	4.5
Total	67	100.0

In terms of medication use to treat the skin changes, 46.3% of the participants reported using medication, while 53.7% did not use any medication.[Figure 4] The onset of skin changes during pregnancy varied, with 50.7% reporting the second trimester as the onset period, followed by the first trimester (26.9%) and the third trimester (22.4%).[Figure 5]

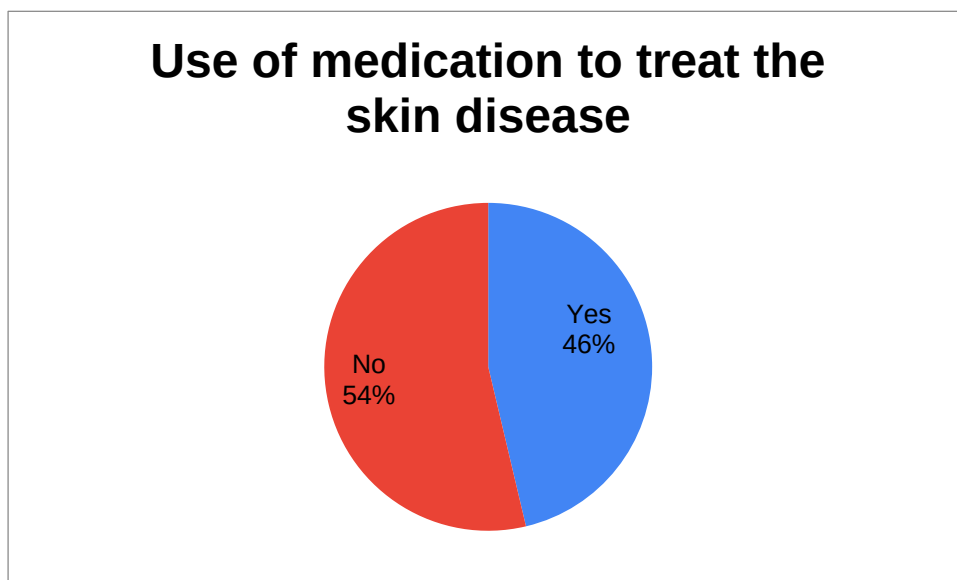


Figure 4 : medication use to treat the skin changes

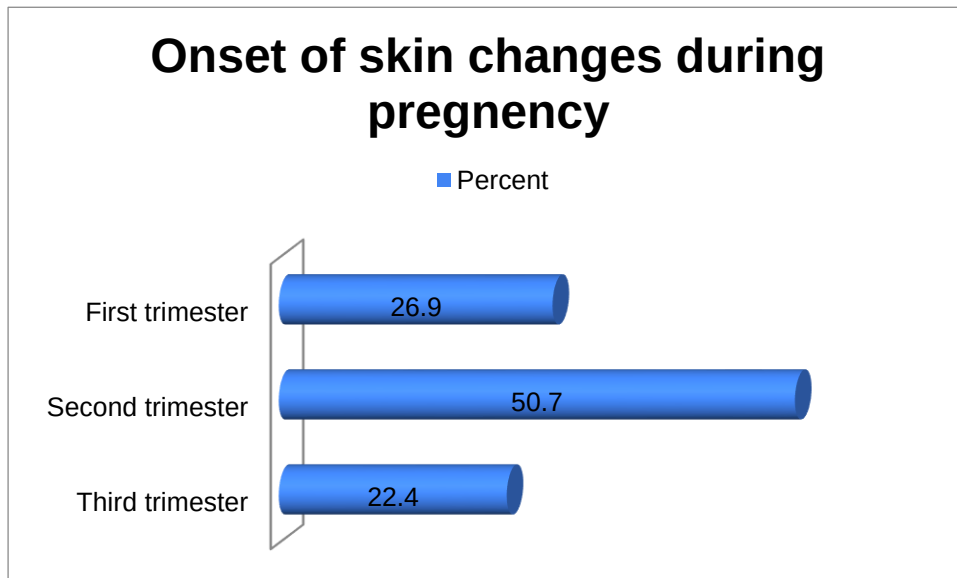


Figure 5: The onset of skin changes during pregnancy

The spread of skin changes across the body was reported by 61 participants. The distribution of affected areas included hands (18.0%), neck (18.0%), abdomen (16.4%), back (4.9%), and face (1.6%), among others.[Table 5]

Table 5: The distribution of affected area

Spread of skin in body	Frequency	Percent
Limited	4	4.9
All body	1	3.2
Stomach	1	1.6
Hands and Abdomen	3	4.9
Hands	11	18.0
Face	1	1.6
Face and hands	1	1.6
Extremities	1	1.6
Back	3	4.9
Neck and hands	3	4.9
Neck and hands and Abdomen	1	1.6

Neck	11	18
Head	7	11.4
Abdomen and joints	1	1.6
Abdomen and back	1	1.6
abdomen	10	16.4
One leg	1	1.6
Chest and back	1	1.6
Total	61	100.0

Among the participants, 42 reported having a chronic disease. The most common chronic diseases reported included gestational diabetes (33.3%), hypertension (9.5%), hyperthyroidism (7.1%), and hypothyroidism (4.8%), and others.[Table 6]

Table 6: Chronic disease among participants

Chronic disease	Frequency	Percent
Hyperthyroidism	3	7.1
No	7	16.7
Sickle cell anemia	1	2.4
Gestational hypertension	2	4.8
Hypertension	4	9.5

Hypothyroidism	2	4.8
Allegers	7	16.7
Gestational diabetics	14	33.3
Asthma	1	2.4
previous stroke	1	2.4
Total	42	100.0

Regarding the effect of skin diseases on participants' lives, 52.2% reported no effect, 35.8% reported a mild effect, and 11.9% reported a moderate effect.[Figure 6]

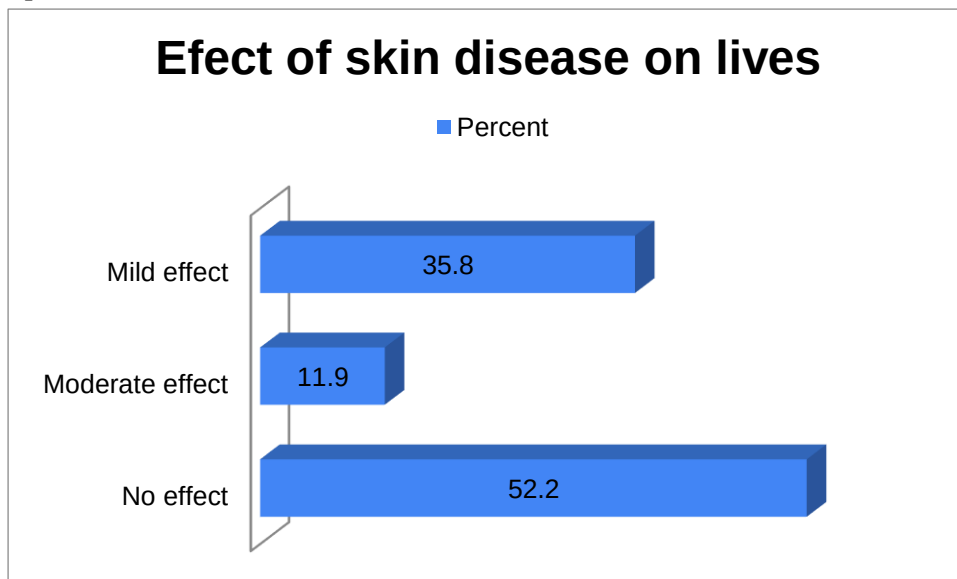


Figure 6: the effect of skin diseases on participants' lives

In terms of the participants' history of previous skin diseases, 30.1% reported having had a previous skin disease, while 69.9% did not have such a history.[Table 7]

Table 7: **History of previous skin disease among participants**

History of previous skin disease	Frequency	Percent
Yes	28	30.1
No	65	69.9
Total	93	100.0

CHAPTER FOUR

4.1 Discussion

The present study aimed to determine the prevalence and pattern of dermatosis among pregnant women in the White Nile State during the period from 2023 to 2024. The study collected data from 100 participants and analyzed various factors related to dermatosis during pregnancy. In order to provide a comprehensive discussion, the findings of the present study will be compared and contrasted with previous studies conducted in different regions.

In terms of age distribution among the participants, the majority fell within the 19-29 age group (45.0%), followed by the 30-40 age group (40.0%). This distribution is consistent with the findings of previous studies conducted by Lt Col MdHumayunKabir et al and OlusolaOlabisiAyanlowo et al, which also reported a higher prevalence of dermatosis during pregnancy among women in their twenties and thirties (Kabir et al., year; Ayanlowo et al., year). However, it should be noted that the present study had a smaller percentage of participants in the older age group of 41-50 (6.0%) compared to the previous studies. This difference may be attributed to the specific demographic characteristics of the White Nile State population.

Regarding the presence of skin changes during pregnancy, the present study found that 69.0% of the participants experienced skin changes, while 31.0% did not. These findings are consistent with the study conducted by MariuszSikora, Ciechanowicz et al, which reported that 74.78% of pregnant women experienced skin changes (Sikora, Ciechanowicz et al., year). However, the prevalence of skin changes in the present study is higher than the findings of Lt Col MdHumayunKabir et al, who reported a prevalence of 6.9% (Kabir et al., year). This difference could be attributed to variations in the study population, geographical location, and methodology.

The specific types of skin diseases observed in the present study included psoriasis, allergy, alopecia, vitiligo, stretching marks, eczema, and varicose veins. These findings are consistent with previous studies, such as the study conducted by Alvi RU, which reported the occurrence of prurigo of pregnancy, dermatoses associated with intrahepatic cholestasis of pregnancy (ICP), polymorphic eruption of pregnancy, and pruritic folliculitis of pregnancy (Alvi, year). It is noteworthy that the prevalence of these specific dermatoses in the present study (15.0%) is higher than the findings of Alvi RU (2.5% for prurigo of pregnancy, 2.0% for dermatoses associated with ICP, 1.5% for polymorphic eruption of pregnancy, and 0.5% for pruritic folliculitis of pregnancy). This discrepancy may be attributed to

the difference in sample size and regional variations.

The onset of skin changes during pregnancy varied among the participants, with the highest percentage (50.7%) experiencing skin changes in the second trimester. This finding is consistent with a previous study conducted by Lt Col MdHumayunKabir et al, which reported that 59.48% of patients experienced skin changes in the third trimester (Kabir et al., year). The present study did not find a significant difference in the occurrence of skin changes based on the trimester, unlike the study by Fernandes LB, Amaral WN, which reported a higher prevalence of physiological changes in the third trimester (Fernandes, Amaral, year). This difference could be attributed to variations in sample size and methodology.

The effect of skin diseases on the quality of life of pregnant women was assessed in the present study. The majority of participants (52.2%) reported no effect on their lives, while 35.8% reported a mild effect and 11.9% reported a moderate effect. These findings are consistent with the study conducted by Alvi RU, which reported no significant impact on the quality of life of pregnant women with specific dermatoses (Alvi, year). It is important to note that the present study did not include a detailed assessment of the specific aspects of quality of life affected by the skin diseases, and further research is needed in this area.

4.2 Conclusion

This study sheds light on the prevalence and pattern of dermatosis among pregnant women in the White Nile State. The findings demonstrate that a significant proportion of women experience skin changes during pregnancy, with various dermatoses observed. The age distribution of participants aligns with previous studies, although the study population in the White Nile State had a smaller representation of older women. The prevalence of skin changes in this study is higher than in some previous studies, highlighting potential regional variations. The most common trimester for the onset of skin changes was the second trimester. Overall, the impact on the quality of life was reported to be mild or negligible for the majority of participants. Further research is needed to explore the specific aspects of quality of life affected and to develop effective management strategies for pregnant women with dermatoses

4.3 Recommendations

Based on the findings of this study on the prevalence and pattern of dermatosis among pregnant women in the White Nile State, the following recommendations are proposed:

- Healthcare providers and professionals should be aware of the high prevalence of skin changes during pregnancy and the various types of dermatoses that can occur. This knowledge will enable them to provide better care and support to pregnant women experiencing these skin conditions.
- Further research should be conducted to explore the specific causes and risk factors associated with dermatoses during pregnancy in the White Nile State. Understanding these factors can help identify preventive measures and develop targeted interventions.
- Healthcare facilities in the White Nile State should establish specialized dermatology clinics or units that cater to the unique needs of pregnant women. These clinics can provide comprehensive care, including diagnosis, treatment, and management of dermatoses during pregnancy.

4.4 Limitation of the study

1. Limited access to healthcare facilities: The ongoing armed conflict in Sudan, , may have hindered participants' access to healthcare facilities. This limitation could have affected the sample size and generalizability of the study's findings. Pregnant women affected by the conflict may have faced challenges in seeking medical care, leading to underrepresentation or exclusion from the study.
2. Disruption of healthcare infrastructure: The armed conflict may have resulted in the disruption of healthcare infrastructure, including dermatology clinics and specialized services. This limitation could have impacted the availability of resources and expertise needed to accurately diagnose and treat

dermatoses during pregnancy. It may have also affected the quality and continuity of care provided to pregnant women in the study.

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5.2 Appendix :

Research Questionnaire :

1-Do you agree to participate in this research?

☐ Yes

☐ No

2-Age:

☐ >18

☐ 19-29

☐ 30-40

☐ 41-50

☐ ≥/≤51

3-Did you have previous pregnancy?

☐ Yes

☐ No

4-If Yes, How many times?

☐ One

☐ Two

☐ Three

☐ More

5- Are you diagnosed with dermatological condition?

☐ Yes

☐ No

6-If yes, what is it?

7- Are you diagnosed with a chronic disease?

☐ Diabetes

☐ Thyroid diseases

☐ Atopy or allergies

8- Are you suffering from dermatological changes in this pregnancy?

☐ Yes

☐ No

9- If yes, How do you describe this dermatological change?

☐ Itching

☐ Redness

☐ Dark spots

☐ Papule or rash

☐ Pigmentations

☐ Others

10-If any, did you sought medical advise in this regard?

☐ Yes

☐ No

11- In which trimester did the change begin?

☐ First trimester

☐ Second trimester

Third trimester

12- What is the extent of this change in the body?

13-Did you suffer from dermatological changes in previous pregnancies?

Yes

No